

Supplementary Material for Minority Legislators (Co)Sponsor Differently From White Legislators: Causal Evidence from U.S. Congress

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1 Information on Treated Congressional District Representatives

Candidate and electoral race information was obtained from Congressional district and candidate profiles across Congress.gov and Ballotpedia.

1.1 Treatment Group

CA24

Democratic U.S. House Representative Lois Capps (White) was California's 24th congressional district incumbent from the start of the study in 2013 all the way to the 2016 elections, until she decided she would no longer for run for re-election and was replaced by Democratic candidate Salud Carbajal (Latine) after he beat Republican candidate Justin Fareed with a 53.4% vote. This election cycle only brought a racial turnover, as the seat went from a white legislator to a black legislator.

DE01

Democratic U.S. House Representative John Carney Jr. (White) was Delaware's congressional district incumbent all the way up to the 2016 elections, until he decided he would no longer for run for re-election and was replaced by Democratic candidate Lisa Blunt Rochester (Black) after she beat Republican candidate Hans Reigle with a 55.5% vote. This election cycle only brought a racial turnover, as the seat went from a white legislator to a black legislator.

CO02

Democratic U.S. House Representative Jared Polis (White) was Colorado's 2nd congressional district incumbent all the way to the 2018 congressional elections, where he no longer ran for office and was replaced by Democratic candidate Joe Neguse (Black) after beating Republican Challenger Peter Yu with a 60.3% vote. This election cycle only brought a racial turnover, as the seat went from a white legislator to a black legislator.

CT05

Democratic U.S. House Representative Elizabeth Esty (White) was Connecticut's 5th congressional district incumbent up to 2018, where she decided not to run for reelection and was replaced

by Democratic candidate Jahana Hayes (Black) after beating Republican Challenger Manny Santos with a 55.59% vote. This election cycle only brought a racial turnover, as the seat went from a white legislator to a black legislator.

MA07

Democratic U.S. House Representative Michael Capuano (White) was Massachusetts' 7th congressional district incumbent all the way up to 2018, where he was then beaten by Democratic candidate Ayanna Pressley (Black) during the Democratic Primaries with a 58.6% vote. Pressley then, ran unopposed and won with a 98.3% vote. This election cycle only brought forth a racial turnover.

NY14

Democratic U.S. House Representative Joseph Crowley (White) was New York's 14th congressional district incumbent all the way to 2018, where he was then beaten by Democratic candidate Alexandria Ocasio-Cortez (Latinx) with a 56.7% vote during the democratic primary election and once again during the general election with a 78.2% vote. This election cycle brought forth only a racial turnover.

TX16

Democratic U.S. House Representative Beto O'Rourke (White) was Texas' 16th congressional district incumbent all the way up to 2018, where he decided not to run and was then replaced by Democratic candidate Veronica Escobar (Latine) after beating Republican candidate Rick Seberger and Independent candidate Ben Mendoza with a 68.5% vote. This election cycle brought forth only a racial turnover.

TX29

Democratic U.S. House Representative Raymond Eugene 'Gene' Greene (White) was Texas' 29th congressional district incumbent from 1993 all the way to 2018, where he decided not to run and was then replaced by Democratic candidate Sylvia Garcia (Latine). She beat Republican candidate Phillip Aronoff with a 75.1% vote. This election cycle brought forth only a racial turnover.

WA07

Democratic U.S. House Representative Jim McDermott (White) was Washington's 7th congressional district incumbent until the 2016 elections when he decided not to run for re-election. He was succeeded by Democratic candidate Pramila Jayapal (Asian), who won the seat by defeating Democratic candidate Brady Walkinshaw with 42.1% of the vote during the primaries, and with 56.1% of the vote during the general elections. This election marked a racial turnover, as the seat transitioned from a White legislator to an Asian legislator.

FL07

Republican U.S. House Representative John Mica (White) served as Florida's 7th congressional district incumbent until the 2016 elections, when he was defeated by Democratic candidate Stephanie Murphy (Asian), who won the seat with 51.5% of the vote. This election cycle brought both a partisan and racial turnover, as the seat transitioned from a White Republican to an Asian Democrat. Representative Murphy had no prior legislative office experience and served until 2022.

FL10

Republican U.S. House Representative Daniel Webster (White) was Florida's 10th congressional district incumbent until the 2016 elections, when he vacated the seat to run in a different district. He was succeeded by Democratic candidate Val Demings (Black), who defeated Republican candidate Thuy Lowe with 64.9% of the vote. This election marked both a partisan and racial turnover, transitioning from a White Republican to a Black Democrat.

VA04

Republican U.S. House Representative Randy Forbes (White) was Virginia's 4th congressional district incumbent until the 2016 elections, when he chose to run in a different district. He was succeeded by Democratic candidate A. Donald McEachin (Black), who defeated Republican candidate Mike Wade with 57.4% of the vote. This election resulted in both a partisan and racial turnover, as the seat went from a White Republican to a Black Democrat.

CA21

Republican U.S. House Representative David Valadao (White) served as California's 21st con-

gressional district incumbent until the 2018 elections, when he was defeated by Democratic candidate TJ Cox (Asian). Cox won the seat with 50.4% of the vote, marking both a partisan and racial turnover. Representative Cox had prior experience as an engineer and small business owner and served one term before losing re-election in 2020.

CA39

Republican U.S. House Representative Ed Royce (White) served as California's 39th congressional district incumbent until the 2018 elections, when he chose not to seek re-election. He was succeeded by Democratic candidate Gilbert Cisneros (Latine), who defeated Republican candidate Young Kim with 51.6% of the vote. This election cycle brought both a partisan and racial turnover, transitioning from a White Republican to a Latine Democrat. Representative Cisneros had no prior legislative office experience and served one term before losing re-election in 2020.

IL14

Republican U.S. House Representative Randy Hultgren (White) was Illinois's 14th congressional district incumbent until the 2018 elections, when he was defeated by Democratic candidate Lauren Underwood (Black). Underwood won the seat with 52.5% of the vote, marking both a partisan and racial turnover. Representative Underwood, a registered nurse and former senior advisor in the U.S. Department of Health and Human Services, has remained an incumbent since.

TX32

Republican U.S. House Representative Pete Sessions (White) served as Texas's 32nd congressional district incumbent until the 2018 elections, when he was defeated by Democratic candidate Colin Allred (Black). Allred won the seat with 52.3% of the vote, marking both a partisan and racial turnover.

KS03

Republican U.S. House Representative Kevin Yoder (White) served as Kansas's 3rd congressional district incumbent until the 2018 elections, when he was defeated by Democratic candidate Sharice Davids (Native American). Davids won the seat with 53.6% of the vote, marking both a partisan and racial turnover. Representative Davids, a member of the Ho-Chunk Nation and former

attorney, has remained an incumbent since.

NM02

Republican U.S. House Representative Steve Pearce (White) served as New Mexico's 2nd congressional district incumbent until the 2018 elections, when he chose to run for governor. He was succeeded by Democratic candidate Xochitl Torres Small (Latine), who won the seat with 50.9% of the vote. This election cycle brought both a partisan and racial turnover, transitioning from a White Republican to a Latine Democrat. Representative Torres Small, a former attorney and field representative, served one term before losing re-election in 2020.

1.2 Placebo Group A — Non-minority Legislators

AZ01

Democratic U.S. House Representative Ann Kirkpatrick (White) was Arizona's 1st congressional district incumbent up to the 2016 elections, until she decided she would no longer for run for re-election and was replaced by Democratic candidate Tom O'Halleran (White) after he beat Republican candidate Paul Babeu with a 50.7% vote. Representative O'Halleran had been an Arizona state representative and senator from 2001 to 2009 and remained an incumbent until the 2022 elections, where he lost his seat.

CA20

Democratic U.S. House Representative Sam Farr (White) was California's 20th congressional district incumbent all the way up to the 2016 elections, until he decided he would no longer for run for re-election and was replaced by Democratic candidate Jimmy Panetta (White) after he beat Republican candidate Casey Lucius with a 70.8% vote. Representative Panetta had no prior legislative office experience and was an incumbent until 2022.

FL09

Democratic U.S. House Representative Alan Grayson (White) was Florida's 9th congressional district incumbent all the way up to the 2016 elections, until he decided he would no longer for run for re-election and was replaced by Democratic candidate Darren Soto (White) after he beat

Republican candidate Wayne Liebnitzky with a 57.5% vote. Representative Soto had served as a Florida state representative and senator for nine years prior to being in office.

MD08

Democratic U.S. House Representative Chris Van Hollen (White) was Maryland's 8th congressional district incumbent all the way to the 2016 congressional elections, where he no longer ran for office and was replaced by Democratic candidate Jamie Raskin (White) after beating Republican Challenger Dan Cox with a 60.6% vote. Representative Raskin had previously served in the Maryland state Senate for nine years.

NY03

Democratic U.S. House Representative Steve Israel (White) was New York's 3rd congressional district incumbent up to 2016, where he decided not to run for reelection and was replaced by Democratic candidate Tom Suozzi (White) after beating Republican Challenger Jack Martins with a 53% vote. Representative Israel had no prior legislative office experience and has remained an incumbent until 2022.

NY25

Democratic U.S. House Representative Louise Slaughter (White) was New York's 25th congressional district incumbent all the way up to 2018, where she passed away and was then replaced by Democratic candidate Joseph Morelle during the 2018 elections with a 59% vote against Republican candidate James Maxwell. Morelle had no prior legislative experience before coming to office.

AZ09

Democratic U.S. House Representative Krysten Sinema (White) was Arizona's 9th congressional district incumbent all the way to 2018, where she decided not to run and was then replaced by Democratic candidate Greg Stanton after beating Steve Ferrara with a 61% vote during the general elections. Representative Stanton had no prior legislative office experience and remained an incumbent until 2022.

MA03

Democratic U.S. House Representative Niki Tsongas (White) was Massachusetts' 3rd congressional district incumbent all the way up to 2018, where she decided not to run and was then replaced by Democratic candidate Lori Trahan (White) after beating Republican candidate Rick Green with a 62% vote.

MD06

Democratic U.S. House Representative John Delaney (White) was Maryland's 6th congressional district incumbent up to 2018, where he decided not to run and was then replaced by Democratic candidate David Trone (White) after beating Republican candidate Amie Hoeber with a 59% vote.

1.3 Placebo Group B — Minority Legislators

CA17

Democratic U.S. House Representative Xavier Becerra (Latine) was California's 34th congressional district incumbent up until 2017 where he was appointed as California's Attorney General. As a result, the district had a special election where out of the top-two candidates, Jimmy Gomez (Dem - Latine) and Robert Lee Ahn (Dem - Asian), Jimmy Gomez won with a 59.2% vote. Representative Gomez served as a California State Assembly member from 2012 to 2017, and remains an incumbent throughout the study period.

CA46

Democratic U.S. House Representative Loretta Sanchez (Latine) was California's 46th congressional district incumbent up to the 2016 elections, until she decided not to run for re-election and run for the Senate instead. The position was filled by Democratic candidate Lou Correa (Latine) after he beat Democratic candidate Bao Nguyen (Asian) with a 70% vote. Representative Correa served as a California State Assembly member and State senator prior to being in office, and remains an incumbent throughout the study period.

FL05

Democratic U.S. House Representative Corrine Brown (Black) was Florida's 5th congressional

district incumbent up to 2016, where she lost to Democratic candidate Alfred Lawson (Black) in the state primaries with a 47.6% to 39% vote. Democratic candidate Lawson then went on and beat Republican candidate Gloreatha Scurry-Smith (Black) in the general elections with a 64.2% vote. Representative Lawson had served as a Florida State Assembly member and senator prior to his position, and remains an incumbent throughout the study period.

IL04

Democratic U.S. House Representative Luis Gutierrez (Latine) was Illinois' 4th congressional district incumbent up to 2016, where he decided not to run for re-election. In the general election, Democratic candidate Jesus Garcia (Latine) defeated Republican candidate Mark Wayne Lorch (White) with a 86.6% vote. Representative Garcia had previously served in the Illinois State Senate, and remains an incumbent throughout the study period.

MD04

Democratic U.S. House Representative Donna Edwards (Black) was Maryland's 4th congressional district incumbent up to 2016, where she decided not to run for re-election and ran for the Senate. In the general election, Democratic candidate Anthony Brown (Black) defeated Republican candidate George McDermott (White) with a 74.1% vote. Representative Brown previously served in Maryland's House of Delegates and remains an incumbent throughout the study period.

MN05

Democratic U.S. House Representative Keith Ellison (Black) was Minnesota's 5th congressional district incumbent up to 2018, where he decided not to run for re-election and became the Attorney General of Minnesota. In the general election, Democratic candidate Ilhan Omar (Black) defeated Republican candidate Jennifer Zielinski (White) with a 78% vote. Representative Omar had no previous legislative experience, and remains an incumbent throughout the study period.

NY13

Democratic U.S. House Representative Charles Rangel (Latine) was New York's 13th congressional district incumbent up to 2016, where he decided not to run for re-election. In the general

election, Democratic candidate Adriano Espaillat (Latine) defeated Republican candidate Tony Evans (Black) with a 88.8% vote. Representative Espaillat had previously served in New York’s Assembly and Senate, and remains an incumbent throughout the study period.

TX15

Democratic U.S. House Representative Ruben Hinojosa (Latine) was Texas’ 15th congressional district incumbent up to 2016, where he decided not to run for re-election. In the general election, Democratic candidate Vicente Gonzalez (Latine) defeated Republican candidate Tim Westley (Black) with a 57.3% vote. Representative Gonzalez had no previous legislative experience, and remains an incumbent throughout the study period.

2 SDID Unit and Time Weight Calculations

As a way to avoid having to arbitrarily select control cases, the SDID estimator incorporates the use of unit ($\hat{\omega}_{sdid}$) and time ($\hat{\lambda}_t^{sdid}$) weights in order to construct a counterfactual that roughly matches pre-treatment trends of unexposed units with the exposed ones and balances both pre- and post-exposure time periods for unexposed units, respectively (Arkhangelsky et al. 2021). As explained in Arkhangelsky et al. (2021), both $\hat{\omega}_{sdid}$ and $\hat{\lambda}_t^{sdid}$ are chosen by the estimator through two calculations.

Unit-specific weights are chosen via a two-step process. First, the estimator computes the optimal regularization parameter (ζ) to match the size of an average single period outcome change (Δ_{it}) for all control units in the pre-treatment period, which is then multiplied by, what the authors call, a theoretically motivated scaling $(N_{tr}T_{post})^{1/4}$.¹ This parameter can be calculated as:

¹In machine learning, regularization is a technique that is used to properly calibrate the adjusted loss function of a model with the goal of preventing both over- and under-fitting.

$$\begin{aligned}
\zeta &= (N_{tr} T_{post})^{1/4} \hat{\sigma} \\
\text{where } \hat{\sigma}^2 &= \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1} (\Delta_{it} - \bar{\Delta})^2, \\
\Delta_{it} &= Y_{i(t+1)} - Y_{it} \\
\text{and } \bar{\Delta} &= \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1}
\end{aligned} \tag{1}$$

where

$$\hat{\sigma}^2 = \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1} (\Delta_{it} - \bar{\Delta})^2 \tag{2}$$

,

$$\Delta_{it} = Y_{i(t+1)} - Y_{it} \tag{3}$$

and

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre} - 1)} \sum_{i=1}^{N_{co}} \sum_{t=1}^{T_{pre}-1} \Delta_{it} \tag{4}$$

Second, the regularization parameter is then implemented to the following objective function:

$$(\hat{\omega}_0, \hat{\omega}^{sdi}) = \arg \min_{\omega_0 \in \mathbb{R}, \omega \in \Omega} \ell_{unit}(\omega_o, \omega) \tag{5}$$

where

$$\ell_{unit}(\omega_o, \omega) = \sum_{t=1}^{T_{pre}} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{it} - \frac{1}{N_{tr}} \sum_{i=N_{co}+1}^N Y_{it} \right)^2 + \zeta^2 T_{pre} \|\omega\|_2^2, \text{ and}$$

$$\Omega = \{\omega \in \mathbb{R}_+^N : \sum_{i=1}^{N_{co}} \omega_i = 1, \omega_i = N_{tr}^{-1} \text{ for all } i = N_{co} + 1, \dots, N\}$$

²Model information comes straight from Arkhangelsky et al. (2021)

As laid out in both Arkhangelsky et al. (2021) and Clarke et al. (2023), the incorporation of the regularization (ζ) term in this specification follows Doudchenko and Imben’s (2016) strategy to introduce it as a penalty in order to increase weight dispersion—leading to more unique $\hat{\omega}_{sdid}$ values.³ In addition, the incorporation of the Euclidean norm ($\|\omega\|_2$) leads to a vector of non-negative weights and an intercept (ω_0) term that grants more flexibility when matching unexposed pre-trends to the exposed ones. While it’s true that having much flexibility can be dangerous due to potential differences between units, the overall model’s inclusion of unit fixed effects (α_i) absorbs any constant difference between them (Arkhangelsky et al. 2021).

Similar to unit weights, time weights are found by having the estimator minimize the following objective function:

$$(\hat{\lambda}_0, \hat{\lambda}^{sdid}) = \arg \min_{\lambda_0 \in \mathbb{R}, \lambda \in \Lambda} \ell_{time}(\lambda_0, \lambda)$$

where

$$\ell_{time}(\lambda_0, \lambda) = \sum_{i=1}^{N_{co}} \left(\lambda_0 + \sum_{t=1}^{T_{pre}} \lambda_t Y_{it} - \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^T Y_{it} \right)^2 \text{ and}$$

$$\Lambda = \{ \lambda \in \mathbb{R}_+^T : \sum_{t=1}^{T_{pre}} \lambda_t = 1, \lambda_t = T_{post}^{-1} \text{ for all } t = T_{pre} + 1, \dots, T \}$$

These results are then, along with the resolved unit weights, applied to the two-way fixed effects regression shown in the main equation in section 3.3.1. It is important to note, that while it does not explicitly appear in the time weights function, a minuscule regularization penalty is added to ensure the creation of a unique set of weights. More information about why this is done can be found in the appendix section of Arkhangelsky et al. (2021).

3 Variance Estimation and Model Inference

To address the problem of having few treated units to conduct asymptotically valid inference, SDID calculates the variance of point estimates using three approaches: placebo permutations, block bootstrapping, and leaving-one-out (jackknife) (Arkhangelsky et al. 2021). For the context

³More specifically, this is achieved through L2 Regularization.

of this paper, I only discuss block bootstrapping. After showing that the SDID estimator can be asymptotically normal and zero-centered, a point estimate’s (τ) confidence intervals can be constructed with the following function:

$$\tau \in \hat{\tau}^{sdid} \pm z_{\alpha/2} \sqrt{\hat{V}_{\tau}} \quad (6)$$

where $z_{\alpha/2}$ is the inverse normal density function, and $\hat{V}_{\tau}$ is the estimated variance calculated with one of the three approaches.

The block bootstrap approach (Efron 1979) consists of taking a large number (B) of bootstrapped resamples consisting of both treated and control units (N). Across each instance of B , a random sample of units from the total sample of units—in this case, congressional district representatives—is drawn with replacement. If the resample does not consist of only treated or control units, the estimated treatment effects ($\hat{\tau}^{sdid}$) are re-calculated and stored $\hat{\tau}_{(b)}^{sdid}$ (Arkhangelsky et al. 2021). This process is repeated B times, and once the procedure is done, the bootstrap variance is calculated as the variance of all resampled estimates across all iterations. While computationally intensive, this form of variance estimation has demonstrated robust performance alongside the SDID estimator (Arkhangelsky et al. 2021).

4 Staggered Adoption

Because the treatment of minority legislators taking over office occurs at two different periods, I take a staggered adoption approach where the units that start treatment at multiple periods in time are broken into adoption-specific groups that fit the block assignment layout and are then separately applied the SDID estimator (Clarke et al. 2023). The results are then aggregated to a weighted average estimate (ATT), where the weights are assigned based on the relative number of treated units and time periods in each adoption group. The ATT from adoption-specific SDID estimates are calculated as:

$$\widehat{ATT} = \sum_{for \alpha \in A} \frac{T_{post}^{\alpha}}{T_{post}} \times \hat{\tau}_{\alpha}^{sdid} \quad (7)$$

where T_{post}^{α} is the total number of treated units in the sample, T_{post} is the total number of post-treatment periods observed in treated unit sample, and $\hat{\tau}_{\alpha}^{sdid}$ is the estimated average effect of that sample (Clarke et al. 2023). Since there are two congressional sessions where new legislators are introduced, the panel will be split into three sets of two block groups. One set will focus on the treated group whereas the other two sets will focus on the placebo groups (non-minority replacing non-minority and minority replacing minority).

5 Synthetic Difference in Difference Plot Results by Group

The first three subsections discuss the three groups that the Synthetic Difference in Difference (SDID) estimator was applied to: (1) the treatment group where minority legislators replace non-minority ones, (2) the first placebo group where non-minority legislators replace non-minority ones, and (3) the second placebo group where minority legislators replace minority ones. Each of these three subsections breaks down the results by treatment block, given that there are two treatment periods where legislators came into office.

5.1 Treatment Group (Minority Replacing White) - Estimate Plots

Figure 1 displays the average bill sponsorship and cosponsorship trends for all minority treated congressional districts, and their generated synthetic comparison groups. The dotted vertical line separates the pre-treatment period where non-minority legislators were in office and the post-treatment period where minority legislators took over office. The blacked shaded regions represent the time weights ($\hat{\lambda}_t^{sdid}$) assigned across all pre-treatment periods. Panels A and B in the figure depict the number of bills sponsored between non-minority legislators and the minority legislators that replaced them after the 2016 and 2018 elections, respectively. Similarly, panels C and D of the figure plots the number of bills cosponsored between non-minority legislators and the minority

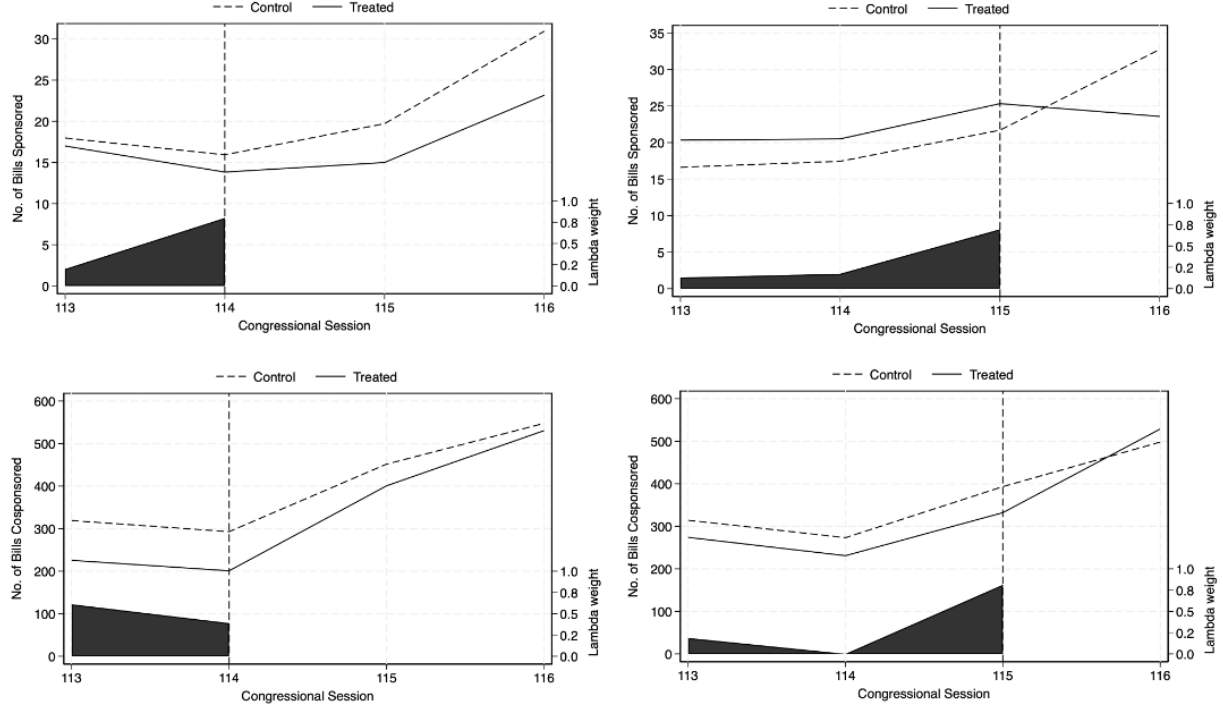


Figure 1: SDID Bill Sponsorship and Cosponsorship Trends (Minority Legislators)

legislators that replaced them after the 2016 and 2018 elections.

A main assumption about SDID is that, without the change in legislator, the non-minority legislators' legislative activity would look like that synthetic control's trajectory. As the results from Figure 1 suggest, the synthetic control groups constructed for all treated group outcomes are nearly parallel to the treated bill sponsorship and cosponsorship trends throughout the pre-treatment periods. Panels A and B show how incoming minority legislators end up sponsoring less bills than their non-minority counterparts would have if they stayed in office. Conversely, panels C and D show incoming minority legislators end up cosponsoring more bills than their non-minority counterparts would have if they stayed in office. Overall, the results across both treatment blocks appear consistent with the direction of the effects for both bill sponsorship and cosponsorship.

5.1.1 Treatment Group (Minority Replacing White) - Unit Weight Plots

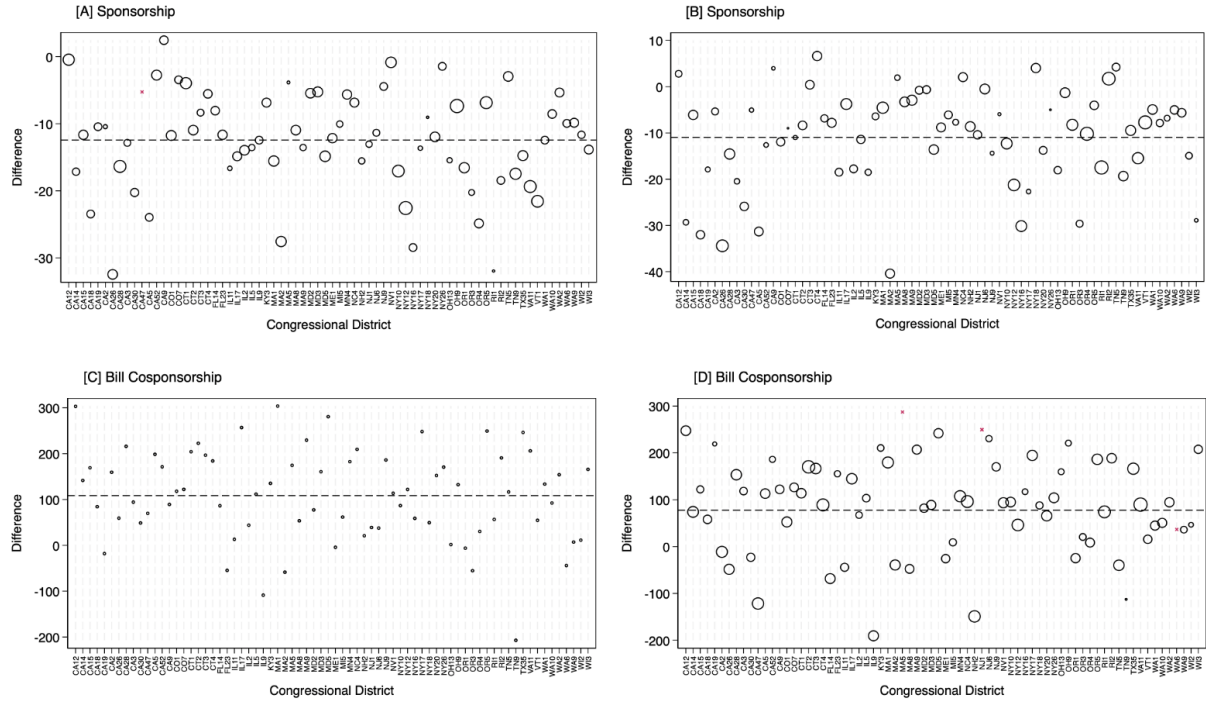


Figure 2: SDID Unit Weights for Treatment Group (Minority Legislators)

5.2 Placebo Group A (White Replacing White) - Estimate Plots

Figure 3 displays the average bill sponsorship and cosponsorship trends for all non-minority treated congressional districts in Placebo Group A, and their generated synthetic comparison groups. The dotted vertical line separates the pre-treatment period where non-minority legislators were in office and the post-treatment period where new non-minority legislators took over office. The blacked shaded regions represent the time weights ($\hat{\lambda}_t^{sdid}$) assigned across all pre-treatment periods.

Like with Figure 1, panels A and B in the figure depict the number of bills sponsored between non-minority legislators and the non-minority legislators that replaced them after the 2016 and 2018 elections, respectively. Similarly, panels C and D of the figure plots the number of bills cosponsored between non-minority legislators and the non-minority legislators that replaced them after the 2016 and 2018 elections.

As the results from Figure 3 suggest, the synthetic control groups constructed for all treated

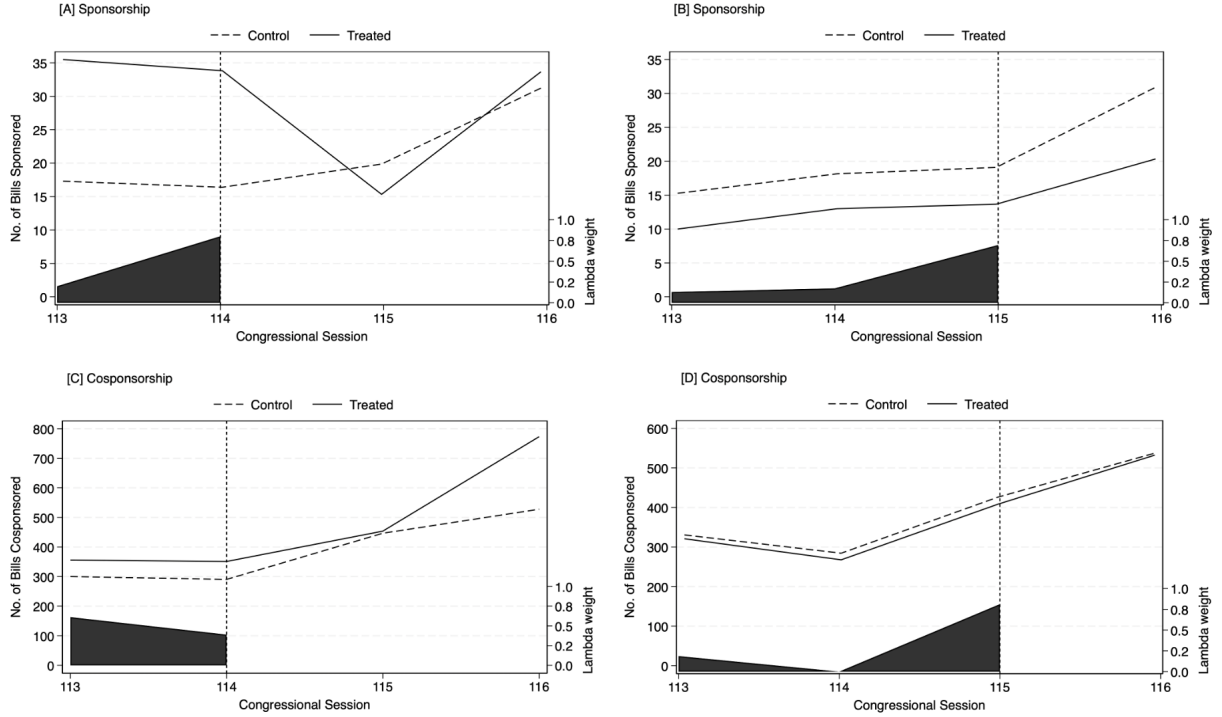


Figure 3: SDID Bill Sponsorship and Cosponsorship Trends (White Legislators)

placebo group outcomes are parallel to the treated trends throughout the pre-treatment periods—implying that the SDID estimator was able to construct a proper comparison group with both the unit and time weights. Unlike the apparent effects of minorities entering office, the effects of non-minorities replacing other non-minorities appear mixed across the two block groups. Panel A shows how incoming non-minority legislators end up sponsoring less bills than their non-minority counterparts would have if they stayed in office. Conversely, panel B shows incoming non-minority legislators end up sponsoring about the same number of bills than their non-minority counterparts would have if they stayed in office.

The results in panel C shows how incoming non-minority legislators do cosponsor more bills than the previous non-minority legislators would have during the 116th (2019-2021). On the other hand, panel D shows hows incoming non-minority legislators do not cosponsor any different from the previous non-minority legislators that were in office. Overall, the results across both treatment blocks appear inconsistent with the direction of the effects for both bill sponsorship and cosponsorship.

5.2.1 Placebo Group A (White Replacing White) - Unit Weight Plots

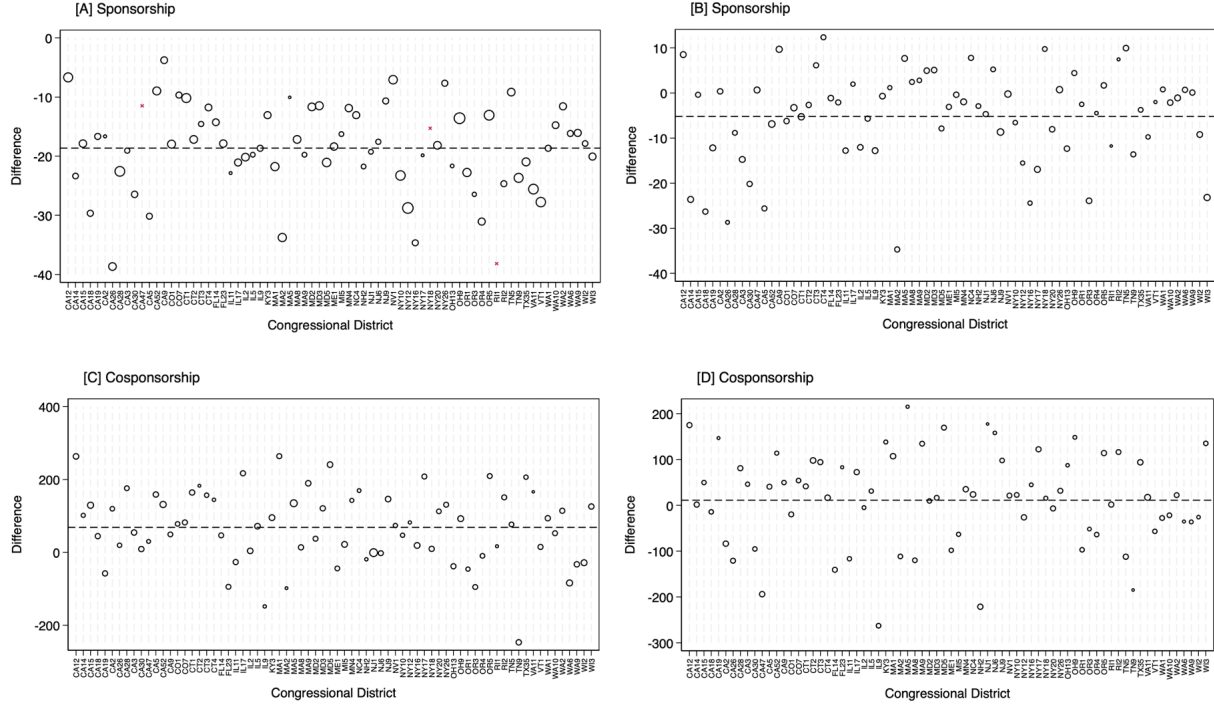


Figure 4: SDID Unit Weights for Treatment Group (White Legislators)

5.3 Placebo Group B (Minority Replacing Minority) - Estimate Plots

Lastly, Figure 5 displays the average bill sponsorship and cosponsorship trends for all treated minority district representatives in Placebo Group B, and their synthetic comparison groups. The dotted vertical line separates the pre-treatment period where minority legislators were in office and the post-treatment period where new minority legislators took over office. The blacked shaded regions represent the time weights ($\hat{\lambda}_t^{sdid}$) assigned across all pre-treatment periods. Like with Figures 1 and 3, panels A and B in the figure depict the number of bills sponsored between minority legislators and the minority legislators that replaced them after the 2016 and 2018 elections, respectively. Similarly, panels C and D of the figure plots the number of bills cosponsored between minority legislators and the minority legislators that replaced them after the 2016 and 2018 elections.

As the results from Figure 5 suggest, the synthetic control groups constructed for all treated

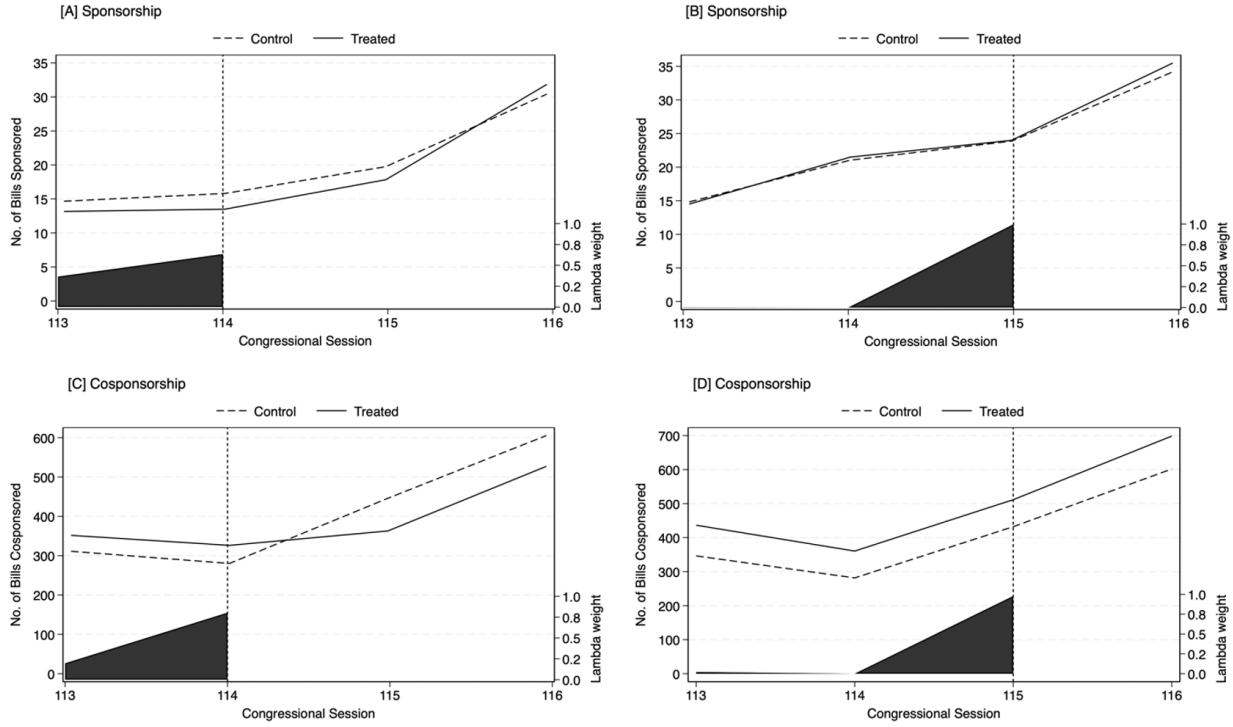


Figure 5: SDID Bill Sponsorship and Cosponsorship Trends (Minority Legislators)

placebo group outcomes are parallel to the treated trends throughout the pre-treatment periods—implying that the SDID estimator was able to construct a proper comparison group with both the unit and time weights. Like with Placebo Group A, the effects of minorities replacing other minorities appear mixed across the two block groups. Panel A shows how incoming minority legislators end up sponsoring less bills than their minority counterparts would have if they stayed in office. Conversely, panels B shows incoming minority legislators end up sponsoring about the same number of bills than their minority counterparts would have if they stayed in office.

The results in panel C shows how incoming minority legislators cosponsor less bills than the previous minority legislators would have. On the other hand, panel D shows how incoming minority legislators do not cosponsor any different from the previous minority legislators that were in office. Overall, the results across both treatment blocks appear inconsistent with the direction of the effects for both bill sponsorship and cosponsorship.

5.3.1 Placebo Group B (Minority Replacing Minority) - Unit Weight Plots

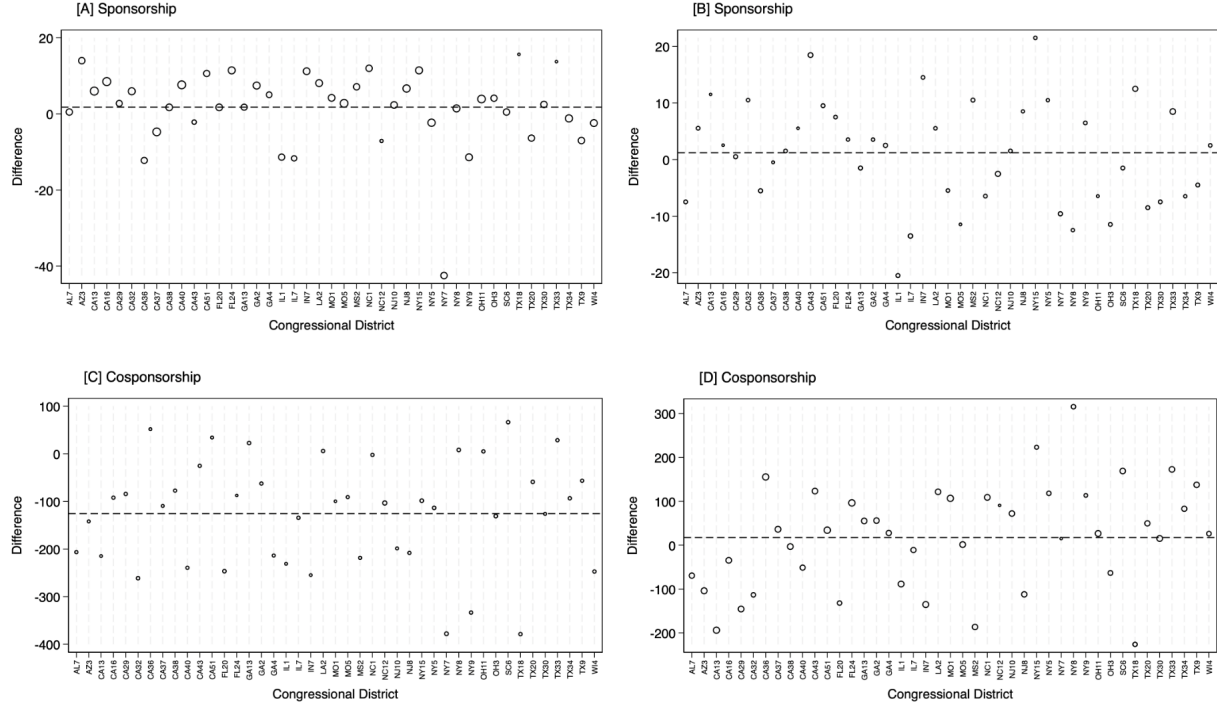


Figure 6: SDID Unit Weights for Treatment Group (Minority Legislators)

6 Mechanism Test

The negative relationship observed in bill sponsorship and cosponsorship for legislators of color could result from several mechanisms. Although the literature highlights the difficulty freshmen face in sponsoring legislation compared to the relative ease of cosponsorship (e.g. Garand and Burke 2006; Hibbing 1991), the results from the placebo tests in the main paper indicate that freshman status does not account for the imbalance in bill sponsorship and cosponsorship among minority legislators replacing White ones. In addition to this mechanism, I explore an additional mechanism: minority legislators' limited sponsorship activity might be attributed to the lack of colleague support stemming from their electoral circumstances, such as primary victories over incumbents. Given the design of the study, it is possible to split the treated congressional districts into groups where these occurrences do happen in order to test if the outcomes of bill sponsorship and cosponsorship for a subset of minority legislators are any different from the main ones.

	(1)	(2)
	Bill Sponsorship	Bill Cosponsorship
Primary Outcomes: Open seat primary	-10.157**	93.623**
	(4.569)	(45.420)
Treated District Representatives	7	7
Observations	316	316
Primary Outcomes: Incumbent beaten in primaries	-18.132**	72.718***
	(9.094)	(20.313)
Treated District Representatives	2	2
Observations	296	296

Note: Bootstrap standard errors in parentheses are based on 1,000 replications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1: SDID weighted estimation results for primary outcomes

6.1 Primary Outcomes

The inverse relationship between bill sponsorship and cosponsorship among legislators of color may be attributed to their position in Congress following primary victories over incumbents, rather than ideological distance. Primary challenges against incumbents are rare at the Congressional level; however, when they do occur, it is plausible that candidates who outprimary opponents might face a lack of support from colleagues as a means of signaling disapproval with electoral outcomes (e.g., Meyer 2021). Given these dynamics, it can be expected that incoming minority legislators who have defeated incumbents would encounter greater obstacles in garnering support for their introduced legislation compared to those who won open-seat districts. Conversely, if primary outcomes rather than marginalization drive this strategic behavior, one might anticipate that incoming minority legislators would exhibit higher levels of sponsorship activity, as they would not face the same level of policy winnowing and other hurdles.

To test this mechanism, I split the nine treated congressional districts into groups where the replacing minority legislator either beat an incumbent in the primaries or won in an open seat district, and run separate Synthetic Difference-in-Differences for each. The results of these tests are displayed in Table 1. Overall, I find no signs of notable changes in the direction of the effects for either bill sponsorship or cosponsorship. These findings suggest that the most plausible mechanisms

of freshman status (as shown in the main paper) or primary election outcomes do not explain minority legislative behavior in sponsorship or cosponsorship, as this mechanism still demonstrates a negative and significant relationship between bill sponsorship and cosponsorship for minority legislators that replace non-minority ones.